Dr. Matthew D. Pressland

November 5, 2026

CONTACT INFORMATION

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RESEARCH INTERESTS

Representation theory of quivers and algebras, categorical and homological methods.
Connections to cluster theory, algebraic and combinatorial geometry.
Abelian, exact and triangulated categories, (higher) Auslander–Reiten theory.
Non-commutative algebra, connections to geometry, mathematical physics, Lie theory.

FELLOWSHIPS

EPSRC Postdoctoral Fellowship, 30.03.20–29.09.23
Originally at University of Leeds, UK, transferred to University of Glasgow, UK, 05.01.22.
Personal grant, value £375,748, funding full-time postdoctoral research position for myself.

Max Planck Institute Guest Programme, Bonn, Germany, 01.04.16–31.03.17
Postdoctoral research fellowship, total value €25,200 plus travel expenses.

EMPLOYMENT

Maître de conférences (≈ lecturer), from 01.12.24
LMNO, **Université de Caen Normandie**, France

Research Associate, 30.09.23–29.09.24
School of Mathematics & Statistics, **University of Glasgow**, UK

Temporary Lecturer, 05.01.22–04.04.22
School of Mathematics & Statistics, **University of Glasgow**, UK

Wissenschaftlicher Mitarbeiter (postdoc), 01.04.17–30.03.20
Institut für Algebra und Zahlentheorie, **Universität Stuttgart**, Germany

EDUCATION

PhD in Mathematics, completed 2015
University of Bath, UK
Adviser: Prof. Alastair King
Thesis title: *Frobenius categorification of cluster algebras*

FURTHER FUNDING

Hausdorff Junior Trimester Programme: New Trends in Representation Theory, €9,300
Funding to lead a three-person research group during the programme.

Foundation Compositio, €6,000
London Mathematical Society Scheme 1 Grant, £5,000
Heilbronn Small Grants Scheme, £4,000
Funding for the conference 'Quivers, Clusters, Moduli and Stability', Oxford, January 2023.

London Mathematical Society Scheme 3 Grant, £1,200
Funding for the research network CLAN (Clusters and their Applications in the North).

Edinburgh Mathematical Society Student Research Bursary, £900
Funding for a student to carry out a summer research project under my supervision.

Heilbronn Research Grant, £7,500
Funded research in a seven-person group led by principal investigator A. King.

PAPERS

13 peer-reviewed publications, 2 preprints and 1 appendix:

16. Quasi-coincidence of cluster structures on positroid varieties
To appear in *J. Eur. Math. Soc. (JEMS)* (published online).

15. Extriangulated ideal quotients, with applications to cluster theory and gentle algebras
With X. Fang, M. Gorsky, Y. Palu, P.-G. Plamondon. *J. Pure Appl. Algebra* **230** (2026), no. 11, Article no. 108370.

14. Cluster categories for completed infinity-gons I: Categorifying triangulations
With İ. Çanakçı, M. Kalck. *J. Lond. Math. Soc.* **111** (2025), Paper no. e70092.

13. Perfect matching modules, dimer partition functions and cluster characters
With İ. Çanakçı, A. King. *Adv. Math* **443** (2024), Paper no. 109570.

12. From frieze patterns to cluster categories
Modern Trends in Algebra and Representation Theory, London Math. Soc. Lecture Note Ser. Vol. 486, CUP (2023), 109–145.

11. A categorification of acyclic principal coefficient cluster algebras
Nagoya Math. J. **252** (2023), 769–809.

10. Calabi–Yau properties of Postnikov diagrams
Forum Math. Sigma **10** (2022), Paper no. e56.

9. On quiver Grassmannians and orbit closures for gen-finite modules
With J. Sauter. *Algebr. Represent. Theory* **25** (2022), no. 2, 413–445.

8. Special tilting modules for algebras with positive dominant dimension
With J. Sauter. *Glasg. Math. J.* **64** (2022), no. 1, 79–105.

7. Mutation of frozen Jacobian algebras
J. Algebra **546** (2020), 236–273.

6. Graded Frobenius cluster categories
With J.E. Grabowski. *Doc. Math.* **23** (2018), 49–76.

5. Internally Calabi–Yau algebras and cluster-tilting objects
Math. Z. **287** (2017), no. 1–2, 555–585.

4. Labelled seeds and the mutation group
With A. King. *Math. Proc. Cambridge Philos. Soc.* **163** (2017), no. 2, 193–217.

3. Cluster structures via representation theory: cluster ensembles, tropical duality, cluster characters and quantisation
With J.E. Grabowski. Preprint, arXiv:2411.11633.

2. Reduction of Frobenius extriangulated categories
With E. Faber, B. Marsh. Preprint, arXiv:2308.16232.

1. Auslander–Reiten translations for Gorenstein algebras
With S. Kvamme. Appendix to *Monomial Gorenstein algebras and the stably Calabi–Yau property*, by A. García Elsener, *Algebr. Represent. Theory* **24** (2021), 1083–1099.

Visiting Positions

Invited participant, Cluster Algebras and Representation Theory, 06.09.21–17.12.21
Isaac Newton Institute for Mathematical Sciences, Cambridge, UK

Group leader, New Trends in Representation Theory, 01.09.20–29.11.20
Hausdorff Institute for Mathematics, Bonn, Germany

Esteem

Awards
Jon Nimmo Teaching Prize for early career staff, Glasgow, 2022

Invitations
Six invited lecture series and many invited conference talks.
Invitations to exclusive conferences at Oberwolfach (×2), Banff International Research Station, Casa Matemática Oaxaca (×3) and Chennai Mathematical Institute.

Main organiser
Noncommutative Geometry Through Categorification, Oaxaca, Mexico, 30.08–03.09.27
Quivers, Clusters, Moduli and Stability 2027, Sānyà, China, 22–26.03.27
ICRA (International Conference on Representations of Algebras), Grenoble, 22.06–03.07.26
Journées d'Algèbre, Caen, 18–19.03.25
Clusters and their Applications in the North, 1st Meeting, Glasgow, 14.02.24
Quivers, Clusters, Moduli and Stability, Oxford, 09–13.01.23
Bridges Between Representation Theory and Algebraic Geometry, Leeds, 24–27.05.22
Representation Theory Working Group, Stuttgart, 2017–18 and 2019–20
Bath Postgraduate Geometry & Algebra Seminar, 2012–15
Bath Postgraduate Seminar Series, 2012–13

Supporting organiser
Algebraic geometry days, Glasgow, 08.05.24–09.05.24
Research network CLAN (Clusters and their Applications in the North), Glasgow organiser (2024–25)
Two Weeks of Silting, Stuttgart, 30.07.19–09.08.19
Calf regional algebraic geometry seminar, Bath, 31.05.12

Refereeing experience
J. Eur. Math. Soc. (JEMS), Proc. Lond. Math. Soc., Adv. Math. (×3), Algebra Number Theory, Int. Math. Res. Not. IMRN, J. Combin. Theory Ser. A, Math. Z., Bull. Lond. Math. Soc., Math. Proc. Cambridge Philos. Soc., J. Alg. (×2), J. Algebraic Combin., Appl. Categ. Structures, Algebr. Represent. Theory (×3), Monatsh. Math., Quart. J. Math. Oxford, Glasg. Math. J., SIGMA Symmetry Integrability Geom. Methods Appl., FPSAC.

External examiner
Eight Bachelors' theses in algebra (three in Norwegian), NTNU Trondheim, 2019

Administrator
FDLIST, mailing list and webpage for representation theory of algebras, with over 400 registered users worldwide, since 2020

Reviewer
MathSciNet, zbMath

Outreach and administration
Invited contributor to 'Snapshots of modern mathematics from Oberwolfach', 2026
Royal Institution Masterclass for secondary school students, Glasgow, 26.11.22
Early Career Researcher roundtable with science minister, on UK R&D Roadmap, 24.09.20
Staff–Student Liaison Committee postgraduate representative, Bath, 2013–14
Teaching Assistant for STEP Outreach Programme for sixth-form students, Bath, 2013

Courses Lectured

Algèbre 1, Université de Caen Normandie, 2026 (combined lecture/tutorial)
Logique et Raisonnement, Université de Caen Normandie, 2026 (combined lecture/tutorial)
Algèbre 1, Université de Caen Normandie, 2025 (combined lecture/tutorial)
Logique et Raisonnement, Université de Caen Normandie, 2025 (combined lecture/tutorial)
Metric Spaces and Basic Topology, University of Glasgow, 2024
Metric Spaces and Basic Topology, University of Glasgow, 2022 (online, Jon Nimmo Teaching Prize)
Höhere Mathematik II, Universität Stuttgart, 2017 (additional lectures)

Supervision

PhD, L. Jeanne, *Positroïdes orthogonaux* (co-supervised with S. Launois), Caen, 09.2026–

MSc Project, D.D. Qin, *The McKay correspondence*, Glasgow, 07–09.2023

L3 Research Internship, K. Marandeau, *Cohen–Macaulay representations*, Caen, 05–07.2025
Summer Project, G.C.-D. Fernández, *Frieze patterns*, Glasgow, 06–07.2022

Talks

Lecture Series
Markov Numbers and the Combinatorics of Cluster Algebras, İstanbul, Turkey, 25–28.08.25
Cluster Algebras and Symplectic Geometry, Banff, Canada, 30.03–04.04.25
Cluster Algebra, Representation Theory and Algebraic Geometry, Hong Kong, 08–12.11.25
Mutations in Representation Theory of Algebras, Online / Isfahan, Iran, 10–12.05.23
Preprojective Algebras and Calabi–Yau Algebras, Online / Ōsaka, Japan, 02–04.03.22
London Mathematical Society Autumn Algebra School, Online, 06–09.10.20

Invited Talks (since 2022)
Journée de la Fédération Normandie-Mathématiques, Caen, France, 17.06.26
Algebra and Representation Theory Seminar, Bonn, Germany, 19.02.26
Séminaire d'Algèbre et Géométries, Grenoble, France, 24.11.25
The RepNet Launch Party, York, UK, 30.07.25
Derived Representation Theory and Triangulated Categories, Thessaloníki, Greece, 27.06.25
Séminaire AG, Versailles, France, 17.06.25
AIMS Conference on Dynamical Systems and Differential Equations, Abu Dhabi, UAE, 18.12.24
Homological Algebra Seminar, Aarhus, Denmark, 23.10.24
Algebra Seminar, Aberdeen, UK, 09.10.24
Séminaire d'Algèbre d'IMJ-PRG, Paris, France, 17.06.24
Séminaire d'Algèbre et de Géométrie, Caen, France, 02.04.24
Séminaire AG, Online / Versailles, France, 19.03.24
London Mathematical Society Northern Regional Meeting, Durham, UK, 25.03.24
Algebraic Topology Seminar, Warwick, UK, 13.03.24
Cluster Algebras and Their Applications, Oberwolfach, Germany, 16.01.24
FD Seminar, Online, 28.09.23
Homological Algebra and Representation Theory, Karlóvasi, Greece, 13.07.23
Nordic Congress of Mathematicians, Aalborg, Denmark, 04.07.23
Combinatorial Aspects of Representation Theory, Oslo, Norway, 28.03.23
Séminaire AG, Versailles, France, 14.03.23
GLEN Algebraic Geometry Seminar, Edinburgh, UK, 09.12.22
Pure Mathematics Seminar, Lancaster, UK, 12.10.22
Geometric and Combinatorial Methods in Homological Algebra, Aarhus, Denmark, 21.07.22
Clusters and Braids Seminar, Online, 06.06.22
Øyvind 60, Trondheim, Norway, 04.06.22
58th ARTIN Meeting, Manchester, UK, 17.03.22
International Workshop on Geometry and Math. Physics, Online / Ulsan, South Korea, 15.02.22
LAGOON Seminar, Online, 10.02.22